

Who Gets a Cure: Venture Capital as Gatekeeper in Biomedical Translation

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Abstract

Venture capital plays a central role in financing early-stage biotechnology development, as its investment criteria shapes therapeutic development patterns. By drawing on historical accounts of the emergence of biotechnology, empirical evidence on early-stage financing, and recent data on venture-backed clinical pipelines, this paper shows that venture capital uniquely supplies the risk tolerance, capital intensity and regulatory coordination required to advance high-risk therapies. However, the concentration of decision-making power in venture capital leads to public health priority mismatch, unequal bargaining power, information asymmetry, and competition conflicts. The paper argues that preserving translational capacity while mitigating these risks requires disclosure of portfolio conflicts and targeted public incentives that steer venture capital toward underfunded areas without undermining its role in translation.

Introduction

Biomedical research routinely produces promising discoveries, but many never reach patients. Between preclinical insight and clinical deployment lies a set of practical, financial, and structural barriers often described in literature as the “valley of death.” The editors of *Nature Reviews Bioengineering* (2023) describe this gap as one that “hinges on a connected ecosystem that includes academics, industry, funding agencies, investors and regulatory bodies.” In other words, discovering a cure is more than just a scientific problem; it is also an operational and financial one. Scientists must design clinical trials, navigate regulatory approval pathways, and plan for market entry years before a therapy can reach patients, a process commonly referred to as biomedical translation. Individual researchers rarely possess the capital, regulatory expertise, trial design, and manufacturing planning, whereas venture capital supplies those capacities through staged financing and control.

The term “gatekeeper” used in this paper is a term coined by Kurt Lewin (1947) and later formalized in network sociology by Ronald Burt (1992). It is used to describe an actor positioned at the junction between a resource and the parties who need it who controls which ideas cross that boundary. In biomedical translation, venture capital occupies this structural position as it determines which scientific discoveries receive sustained investment and which are left behind. Though venture capital may privilege

certain technologies and generate ethical tensions, this paper argues that venture capital functions as the primary enabler of biomedical translation as no alternative funding model provides comparable capital intensity, risk tolerance, and regulatory coordination for clinical translation.

Literature Review

In existing literature, scholars recognize that venture capital plays a major role in bridging the “valley of death” and separately identify ethical tensions in its practice. Scholarship offers less insight into ethical problems arising from venture capital’s gatekeeping power, since investors decide which therapies move forward and which never get the chance. This review spans: (1) venture capital’s rise in biotechnology; (2) why alternative funding mechanisms fall short; (3) venture capital’s structural advantages and selection dynamics; (4) ethical concerns that arise from those dynamics; and (5) the unresolved question that follows.

Venture Capital Relevance to Biotechnology Innovation

Venture capital has been intertwined with biotechnology translation since the inception of the industry. In *Genentech: The Beginnings of Biotech*, science historian Sally Smith Hughes (2013) documented how early VC firms did not merely finance Genentech but actively shaped which scientific discoveries were deemed worth developing, effectively functioning as gatekeepers over what counted as commercially translatable research. As venture capital funding has grown to become the primary bridge between academic science and clinical development, their gatekeeping role has only grown more consequential.

Economists William R. Kerr, Ramana Nanda, and Matthew Rhodes-Kropf (2014) extended this historical analysis by quantifying venture capital’s disproportionate influence on the economy in their widely recognized article, *Entrepreneurship as Experimentation*. They noted that: “as a rough benchmark, about 1000 of the over 500,000 firms founded each year in the United States obtain venture capital finance, but these firms account for about 40 percent of new US publicly-listed companies over the last three decades” (Kerr et al., 13). Their findings prove that a small investor group, venture capital, disproportionately shapes which therapies ever get a chance to become products, despite backing only a fraction of startups that reach public markets.

In biotechnology especially, venture-backed firms dominate the development pipeline. Professor So-Yeon Kang and her coauthors (2024) examined investments between January 2014 and March 2024. Linking proprietary venture capital deal data from Pitchbook to clinical trial records, Kang et al. focused

on investment patterns across drug types, clinical trial phases, and therapeutic areas, identifying 3,749 biomedical venture investments with a median funding amount of \$20 million per deal. Nearly half of identified deals were made to companies that had Phase 1 trials either underway or starting within 24 months of the investment, while only 4.5% were linked to companies with Phase 2 trials exclusively and 0.3% to Phase 3. The concentration of funding in the early stages is notable because Phase 1 represents the stage where therapies face the most risk. During this phase, efficacy remains undemonstrated and regulatory approval is years away; however, substantial capital is simultaneously required to initiate human trials. Venture capital is thus structurally positioned to fund biopharmaceutical development during these early, high-risk clinical stages.

Venture capital's structural positioning is especially visible in the regulatory pathway, where there exists a gap between academic and industry-sponsored trials. Hoffmann and colleagues (2024) found that academic trials faced greater difficulty than industry-sponsored ones at two distinct stages: ethics committee review, where limited staffing slowed submissions, and communication with competent authorities over trial design and manufacturing compliance, where lack of experience led to inadequate documentation. Academic scientists are rarely equipped to manage either bottleneck. Venture-backed companies, by contrast, build regulatory expertise into their financing structure from the start, tying staged milestones to approval thresholds and bringing in consultants before each hurdle arises.

While Kang et al. and Hughes established venture capital's historically significant role in the biotechnology market, neither fully explains why this centrality emerged. Understanding venture capital's necessity requires examining why other potential funding sources like scientific grants and public markets fail to support early-stage biotech development.

Alternative Funding Mechanisms

Pierre Azoulay and Danielle Li (2020) examine grant impact in early biomedical research, arguing that competitive grants work best in environments with large knowledge spillovers and open-ended research problems. Federal agencies like the National Institutes of Health (NIH) fit this model well, prioritizing academic laboratories and student training because these investments generate broad scientific knowledge that benefits the wider research community. However, this same structure is poorly suited for translation. In reality, academic grants reward publications and conference presentations, not the slow, regulated, operational steps of multi-year clinical development that determine whether a promising therapy actually reaches patients.

Kerr, Nanda, and Rhodes-Kropf similarly examined why public markets and big pharma companies cannot serve early stage biomedical innovation. By analyzing firm-level constraints and

investment cycles, they found that public equity markets were too focused on quarterly earnings to handle development timelines that span many years. Likewise, they discussed how internal politics in large pharmaceutical firms—budgeting that favored safe bets and a culture that penalized managers for failed projects—stifled innovation (Kerr et al., 6). They concluded that while these giants were effective at improving existing drugs, their structure fails to support exploring uncertain new science.

Grant funding prioritizes spillovers and scientific knowledge advancement; public markets demand short-term returns; and big pharma is too risk-averse for unproven technologies. When combined, these findings reveal a significant institutional gap: traditional capital lacks the necessary risk tolerance. The constraints of biomedical translation, including long timelines and high failure rates, fail to match the incentive structures of grants, public markets, or established firms. Early clinical translation thus depends on investors willing to tolerate such conditions, which positions venture capital as the key enabler.

Structural Advantages of Venture Capital

Literature characterizes venture capital as resilient relative to other biomedical funding sources. Economists Josh Lerner and Ramana Nanda (2020) demonstrated that venture capital's continual support persists across different technological waves and through various market cycles through historical industry data (Lerner and Nanda, 238). They attributed this consistency to more than just capital. In their view, venture funds also supplied governance, hiring, regulatory consultants, and commercialization planning, rendering venture-backed firms with the operational infrastructure to sustain development through market downturns.

Recent market data reinforces this point. Emily Capra (2023), an editor at McKinsey, analyzed biotech financing between 2021 and 2023 and documents a sharp contraction in public biotech equity markets during this period, which contrasted the comparatively stable venture investing. She attributed venture capital's stable investments to their sensitivity to market signals and ability to reallocate investment, noting a shift toward emerging areas such as machine learning-enabled drug discovery, cell therapies, and precision medicine platforms.

However, such selective reallocation risks narrowing investment to a concentrated set of therapeutic areas. Kang et al. found that within a large dataset of 3,749 biomedical venture investments, 73.8% targeted small-molecule drugs, 12.5% focused on biologics, and only 0.6% supported gene therapies, with 27.8% of investments concentrated in oncology (Kang et al., 2). These categories are not ethically neutral: small molecules and cancer drugs tend to command high prices and serve wealthy-country patient populations, making them more financially attractive than therapies targeting rare

diseases or neglected conditions. When capital clusters this heavily, rare diseases or treatment for disadvantaged communities are deprioritized by projected return.

In plain terms, venture capital decides which ideas receive an early opportunity and the sustained resources necessary to survive. Kerr, Nanda, and Rhodes-Kropf describe this mechanism directly: venture firms fund multiple early-stage projects, discontinue those that underperform as data emerges, and scale the few that demonstrate commercial promise (Kerr et al., 12). For example, after interim Phase 1 data, investors may decide whether to finance a costly Phase 2 trial, pivot the therapy toward a more commercially viable indication, or terminate the program altogether. Selection therefore occurs at two stages—both at initial funding and at subsequent rounds that determine whether a therapy advances toward market.

Gatekeeping at both stages raises critical ethical questions: if venture capital prioritizes therapies based on commercial viability rather than medical need, patients with rare diseases or conditions requiring novel therapeutic approaches may be systematically disadvantaged. When the primary mechanism of biomedical translation allocates resources according to market logic, the resulting therapeutic landscape may reflect profit potential more than public health priorities. Venture capital-led translation therefore demands ethical scrutiny.

Ethical Concerns of Venture Capital Funding

Existing literature underscores a significant tension between the scientific goals of biotechnology and the financial requisites of venture capital.

Lehoux and colleagues (2016) observe directly that venture capital “is not designed to fulfil society's most urgent public health priorities,” a characterization the investment data supports rather than contradicts. The underlying survey data, reported by Ackerly and colleagues (2009), found that up to 85% of capital investors rated public health impact as “not at all or somewhat important” to their investment decisions. The finding describes a mandate, not an indictment of individual investors. Venture capital is structured to generate portfolio returns within a defined time horizon, and public health impact is rarely legible within that scope. The consequence is a structural market failure: when commercial return is the primary selection criterion, therapeutically important but commercially unattractive disease areas go underfunded, as the incentive structure fails to price in that value regardless of investor intent.

Beyond decisions about which ventures to fund, ethical tensions also arise in the criteria and processes through which those investment decisions are made. Through twelve vignettes, professors Yves Fassin and Will Drover analyze conflicts arising from power and information asymmetry. With these stories, they form categories for ethical failures: operational, strategic, governance, relational, and

ownership issues. Fassin and Drover reveal how these ethical failures arise in common venture practices, such as unequal bargaining power and opaque financial arrangements, thereby raising ethical concerns even when those practices are legally permissible. In one vignette case, for example, “one angel investor is pressured to sign a proxy to allow the capital increase to be executed and hence agrees with a serious dilution without having been consulted throughout the negotiations and without having received key documents, which the law requires” (Fassin and Drover, 659). This is an example of how standard financing mechanics can become ethically indefensible even when no law is explicitly broken.

Ethical concerns are not limited to conflicts within a single venture-backed company; rather, they also emerge from the structure of venture firms themselves, which commonly hold stakes in multiple startups simultaneously. This risk is known as “competitive information leakage” (Cox Pahnke et al., 1334). Emily Cox Pahnke, a professor at the University of Washington, and her colleagues document how confidential technical and strategic data can flow between rival startups through “indirect transfers” of shared investor networks and board members (Cox Pahnke et al., 1336). Those in these networks may unintentionally carry insights from one startup to another, suggesting that the venture structure itself can compromise the competitive integrity of the very innovations it claims to support. In life sciences especially, where a single breakthrough can determine whether a treatment reaches patients or stalls in development, this compromise is a public health one.

Gaps and Unresolved Questions

The literature concurs that venture capital is structurally necessary for biomedical translation but accentuates an unresolved tension. Hughes, Azoulay and Li, Kerr et al., and Lerner and Nanda show that venture capital uniquely fills the gap between discovery and clinical viability that grants and large firms cannot. Concurrently, scholars reflect on the ethical concerns of the power venture capital holds over biotechnology innovation. Kang et al. discuss concentration in certain diseases and modalities; Lehoux et al. identify conflicts with public health goals; Fassin and Drover examine power asymmetries; and Cox Pahnke et al. show distorted competition. There exists a paradox here: the traits that make venture capital effective, risk tolerance, staged funding, rapid failure, investor control, also centralize power over which therapies advance.

The literature has not resolved how to reduce the ethical risks of venture capital’s gatekeeping while preserving its translational capacity. This paper therefore argues that, although venture capital inevitably privileges certain biomedical technologies and generates ethical tensions, the solution is developing governance and incentive mechanisms that preserve venture capital’s translational capacity.

Discussion

The literature reveals a central paradox: venture capital is structurally necessary for bridging the valley of death, yet its reliance on market-based selection shapes which therapies ultimately advance. When returns become the gatekeeping criterion for clinical advancement, the therapeutic landscape reflects market optimization rather than disease burden and public health need. Yet no alternative funding model provides the requisite capital intensity, regulatory help, and risk tolerance to move therapies through early clinical stages; eliminating or substantially restricting venture participation would likely deepen the valley of death, rather than bridge it. Consequently, the issue at hand is how its gatekeeping power can be constrained without undermining its translational capacity. Two categories of reform illustrate the possibility: disclosure requirements that make gatekeeping visible, and targeted public incentives—like the Orphan Drug Act and the Priority Review Voucher program—that expand the frontier of commercially viable development.

One way to constrain its gatekeeping power is to introduce layered governance mechanisms that impose accountability and make selection criteria more transparent. Developments in ESG disclosure within biotech suggest that this is feasible. As Forbess and Llewellyn document, ESG reporting among public biotech companies doubled between 2021 and 2022 (Forbess and Llewellyn, 2022). In these, companies identify which board committee oversees ESG, describe data verification strategies, and implement disclosure requirements to ensure consistency across filings. Standards like the SASB Biotechnology & Pharmaceuticals Index, a framework that sets baseline expectations for what companies must report on clinical, ethical, and access-related practices, force companies to report clinical trial practices, business ethics, patient safety, and access to medicines, all areas vital to venture-backed development. Extending this discipline to venture-backed pipelines before IPO is achievable: academic medical centers and venture funds could require similar reporting through term sheets or partnership agreements, embedding disclosure obligations at the point of investment rather than waiting for public markets to impose them. Doing so would require funds to disclose portfolio overlaps, board interlocks, and conflicts between competing firms — precisely the information that renders gatekeeping visible and therefore contestable.

Beyond increasing governance mechanisms, a deeper intervention would focus on restructuring incentives so that treatments with high public health value but low expected returns can attract sustained investment. The Orphan Drug Act of 1983 illustrates the first such mechanism. As Saltonstall, Ross, and Kim explain, Congress addressed rare disease underinvestment by granting seven years of market exclusivity, a substantial tax credit for clinical testing, and federal research grants (Saltonstall et al., 2024). These provisions altered the risk-reward profile of rare disease development. Prior to the Act, only

38 rare disease drugs had received FDA approval; by the end of 2022, 882 drugs had been approved to treat 392 rare conditions, and orphan drugs now account for a substantial share of innovative FDA approvals. The ODA therefore illustrates that venture investment patterns can be steered by adjusting exclusivity, risk, and return without dismantling the market-based model itself. Yet incentives can also create pricing and gaming concerns, so policy design matters: the ODA has faced criticism for enabling firms to repurpose existing drugs for rare indications to capture exclusivity benefits without proportional therapeutic innovation. Extending such mechanisms to other underserved areas would require calibrated eligibility criteria and market pricing review to prevent the same distortions.

The Priority Review Voucher program, established in 2012 for rare pediatric diseases, illustrates a second policy mechanism that reshapes venture opportunity without constraining market logic. Under the program, companies that develop and gain FDA approval for novel therapies treating designated rare pediatric diseases receive a tradeable voucher granting expedited regulatory review for an unrelated drug candidate (National Organization for Rare Disorders, 2025). Critically, the voucher itself becomes a financial asset that companies can sell. Since the program's inception, 63 vouchers have been awarded across 47 rare pediatric diseases, of which only four had any FDA-approved treatments before 2012. Of these diseases, 29 vouchers have been redeemed to date for 26 different drugs. The redeemed vouchers have supported drugs largely outside the Medicare top-spending categories, suggesting the program has steered investment toward genuinely neglected areas rather than blockbuster opportunities. By converting regulatory authority into transferable financial instruments, the PRV program accomplishes what the ODA achieved through market exclusivity and tax credits: it makes commercially unattractive disease areas profitable to develop. The mechanism differs in form but shares the core principle: policy creates new value capture opportunities rather than constraining where venture capital can operate.

Together, these interventions reflect a broader principle: preserving venture capital's translational capacity and addressing its mandate gap are not competing goals. Disclosure of portfolio conflicts makes the gatekeeping function visible and contestable without disrupting the financing structures that make early clinical translation possible. Targeted public incentives, modeled on the ODA and PRV programs, expand the frontier of where venture capital can profitably operate rather than constrain it. The therapeutic landscape will continue to reflect market optimization over disease burden until the incentive structure prices in public health value — and well-designed policy is the mechanism for doing that.

Conclusion

Biomedical research continues to produce promising discoveries, yet crossing the valley of death still depends heavily on venture capital to finance, govern, and coordinate early clinical translation. Grants, public markets, and large pharmaceutical firms rarely provide comparable risk tolerance, capital intensity, or regulatory support at this stage. Venture capital thus shapes which therapies reach the clinic and which remain undeveloped. Because these choices follow market incentives more closely than public health needs, biomedical innovation tends to favor commercially attractive diseases and patient populations over neglected ones. As reliance on private capital intensifies, policy must govern this gatekeeping power with greater precision. Stronger transparency in venture-backed pipelines, disclosure of portfolio conflicts, and targeted incentives for underserved areas would sustain translational investment while directing biomedical development more closely toward public health priorities. How the bridge across the valley of death is structured will help determine not only which therapies survive, but which patients and communities ultimately benefit from scientific progress and which new markets venture capital gets to build.

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